

Papop Lekhapanyaporn

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EXPERIENCE

GNS Systems GmbH | HPC System Engineer Jan 2026 — Present

- Building a high-performance in-house replacement for legacy cluster management software in **C++**.

Redcare Pharmacy | Working student Aug 2023 — Jan 2026

- Built a **Python** monitoring tool that tracked Gematik e-prescription updates and sent automated event notifications, cutting manual tracking effort for the team.
- Built a local **Wiremock/Python** test environment for e-prescription validation; automated template-to-mock conversion to enable early compliance checks.
- Wrote **Python** scripts for bulk e-prescription refactoring — type conversions and field edits — accelerating test-case generation.

EDUCATION

RWTH Aachen University | B.Sc Computer Science Oct 2022 — Sep 2025

- **Thesis**: Benchmarked automated memory management on **NVIDIA GH200**, characterizing performance trade-offs across diverse memory access patterns.

PROJECTS

SPOS — Embedded OS on ATmega664 | C AVR Apr 2024 — Jul 2024

- Built a bare-metal OS for the **ATmega664** in **C/AVR-GCC**, implementing interrupts, critical sections, preemptive task scheduling, and dynamic memory spanning internal SRAM and external **23LC1024** RAM.
- Demonstrated OS capabilities through LED matrix output, joystick input, and a real-time **Snake** game running on the custom scheduler.

SPACE AC | Software Engineer Oct 2020 — Mar 2022

- **SPOROS** | Arduino C Python Qt5 Nov 2020 — Jul 2021
 - Led full-stack embedded development in **Arduino (C)** for two autorotating payloads and a CanSat relay, plus a **Python/Qt5** ground station with real-time telemetry visualization.
 - Designed the inter-payload communication protocol enabling mid-air telemetry relay; secured **3rd place** at the **Annual CanSat Competition 2021**.
- **AlienSat** | Arduino C Python Qt5 Aug 2021 — Feb 2022
 - Developed real-time firmware for a **thermal camera** CanSat payload, streaming temperature arrays for PM2.5–heat correlation analysis.
 - Built live thermal visualization in the **Python/Qt5** ground station and designed the end-to-end payload-to-ground communication protocol.
- **Passenger Balloon** | Arduino C Python Oct 2020 — Mar 2022
 - Flew **three high-altitude balloon missions** deploying CubeSat payloads to **35 km**, using **Arduino** and **Raspberry Pi** for autonomous imaging and atmospheric sensing.
- **Mentoring** | Oct 2021 — Mar 2022
 - Designed and ran a structured onboarding program covering programming fundamentals, project workflow, and the team's hardware/software stack.

EXTRACURRICULAR ACTIVITIES

Interact Club | President, District Vice President May 2020 — Mar 2022

- Led a **Rotary International**-supported student club; organized annual events and drove cross-club community service initiatives.

TECHNICAL SKILLS

Language: Python, C/C++, SQL

Framework: CMake, AVR, Arduino, CUDA, OpenMP, MPI

Tools: Git, Github/Gitlab, RaspberryPI, XCTU(Zigbee)

LANGUAGE

Thai: Native

English: C2

German: C1